

Step-by-Step Derivative

$$f(x) = e^{\sin(x^2)}$$

Step 1: Chain Rule (exp)

$$\frac{d}{dx}[e^{\sin(x^2)}] = 2xe^{\sin(x^2)}\cos(x^2)$$

Step 2: Chain Rule (sin)

$$\frac{d}{dx}[\sin(x^2)] = 2x\cos(x^2)$$

Step 3: Power Rule

$$\frac{d}{dx}[x^2] = 2x$$

Step 4: Simplify

$$\frac{d}{dx}[2xe^{\sin(x^2)}\cos(x^2)] = 2xe^{\sin(x^2)}\cos(x^2)$$

Final Result

$$f'(x) = 2xe^{\sin(x^2)}\cos(x^2)$$